Review of *Hierarchical structuring of weapon allometry in the European earwig*

In this paper the authors explore the scaling relationship between forcep length and body size (pronotum width) in female and male earwigs, under different environmental (nutrition and population density) conditions. The males come in macrolabic and brachylabic morphs, with large and small forceps respectively, and these two forms reflect differences in mating strategy (macrolabic: fighting males; brachylabic: sneaker males). The goal of the research is to determine whether the patterns of scaling across sexes, morphs and environmental conditions reflect changes in the intercept of the mandibles' scaling relationship with body size or the slope of the scaling relationship. The authors argue that this gives insight into the developmental mechanisms that regulates scaling and scaling divergence.

Several authors have explored the pattern of scaling relationships between exaggerated traits and body size, where the trait shows different scaling relationship between sexes and within sexes between males (horn scaling in Onthophagus beetle being the canonical example). What is particularly interesting here, however, is that the two male morphs appear to overlap considerably in overall body size, which is not clearly observed in horned beetles. This indicates that morph identity (and trait exaggeration) is not simply a reflection of overall body size. There are other systems that show this, for example bulb mites, but it appears to be much less common.

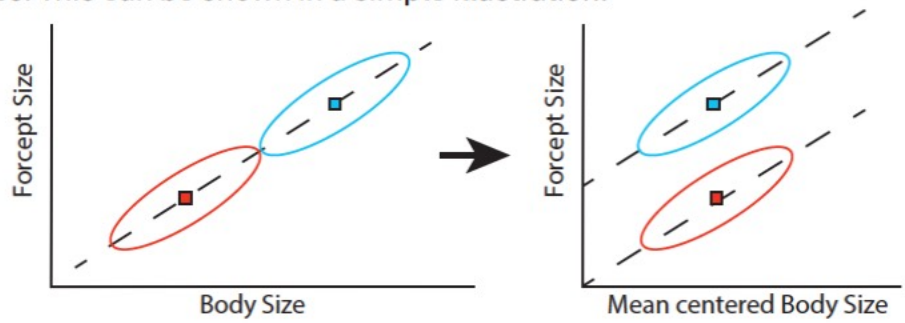
While I think that the data are of value to researchers interested in the evolution of morphology, particularly the morphology of secondary sexual characteristics in males, I have some concerns about the paper.

Comments

1) Much of the results are concerned with differences in the intercept versus slope of the forcep-body scaling relationships among groups (males v. females, high v. low diet, macrolabic v. brachylabic, etc). The authors correctly observed that intercept and slopes in scaling relationships are typically correlated (unless a scaling relationship is rotating around the intercept, any change in slope must also change the intercept). The authors thus center the scaling relationships on mean body size (log pronotum width) per group to "facilitate interpretation of intercepts and reduce collinearity among predictors". This, however, introduces additional interpretation issues that the authors do not address.

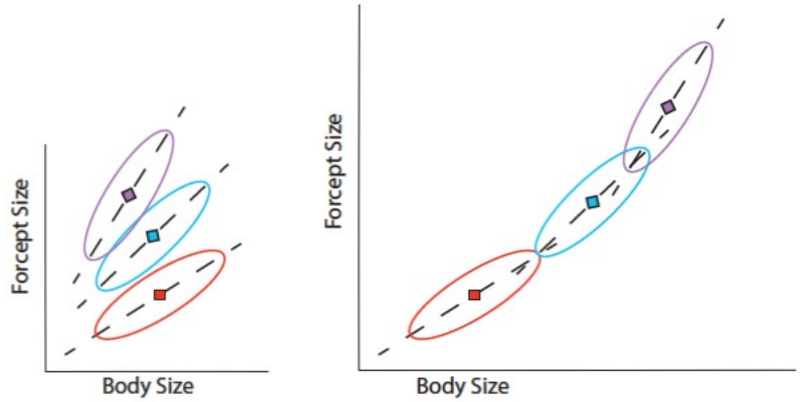
   As an example, consider the effect of diet on forcep size in macrolabic males [L258]. The authors state that "diet *reduced intercept in macrolabic males [and that] Nutritional stress, therefore, disproportionately reduced weapon size in the exaggerated morph".* This interpretation may be an artifact of the method of analysis. If the null hypothesis is that there is a single scaling relationship between body size and forcep size is in macrolabia males, and that 'poor' diet males are

smaller than 'good' diet males, then centering the scaling relationship on mean body size within each group will necessarily result in a lower intercept in the poor diet males. This can be shown in a simple illustration:



Here, red is poor diet, cyan is good diet. The left panel is the raw scaling relationship. The right panel is the scaling relationship when body size in each group is mean-centered. If the slope of the scaling relationship = 1, then there is no change in body proportion, despite an apparent change in intercept for the mean-centered scaling relationships. Thus a change in intercept the mean-centered data does not necessarily mean a change in proportion. It is true that if the slope of the scaling relationship >1, then poor diet males do have disproportionally smaller forceps; however this is captured by the slope of the scaling relationship in the uncorrected data. In short, by (unnecessarily) mean-centering the body size measurements, the authors are obfuscating the true patterns among the scaling relationships. The authors will be able to interpret their data much more easily if they do not mean center the data.

The only analysis that they will not be able to conduct is looking at the correlation between slope and intercept among families, to test whether families with sons that had larger forceps also had steeper scaling relationships. This is because we *a priori* expect slopes and intercepts to be correlated. However, this could be addressed by simply correlating the slope of the family scaling relationship against mean family forcep size (figure below, left). Note, that there would also be a positive correlation if the scaling relationship between forcep and body size were not linear, but curved up (figure below, right), but this can also be tested for explicitly.



2) A related concern is that, by centering groups on body size, then the interpretation of the effects of the environment on the slopes of the scaling relationship are challenging. For example, the [L321] "*size–trait relationship becomes less steep under a poor diet"* in males may be a consequence of non-linear scaling relationship between body size and forcep size, with low diet reducing body size, and hence reducing scaling. (See right panel above). Scaling relationships are a consequence of covariation in trait size as both vary in response to genetic and environmental regulators of size. The authors modify environmental factors that affect size (good diet and high density both increase body size, according to the supplemental table), but do not build this into their explanation of how these factors affect scaling. Both good diet and high density increase body size and correspondingly increase the slope of the scaling relationships: could this not be because the scaling relationship is non-linear?

3) Related to this, the authors report that diet and density effects on slope in males [L321], yet the credibility intervals for all these parameters include zero [L261 onward]. I would argue that there is no reasonable statistical support for this effect.

4) In numerous places the authors over-interpret their data and/or are imprecise with respect to the developmental implications of their findings. This is particularly true concerning interpretation/discussion of intercept versus slope. For example the authors argue that [L30] "*difference in scaling exponent reflect variation in how strongly a trait responds to body size, whereas intercept divergence reflects differences in baseline trait expression at a given body size".* First, implicit in this statement is that trait size is a consequence of body size. However, body size is a composite measure comprising multiple traits. The developmental mechanisms that regulate covariation in trait sizes are complex, but almost certainly not as simple as "body size determines trait size". Second, it is not at all clear what 'baseline' trait expression means developmentally? Baseline relative to what? More generally, authors allude to developmental mechanisms in interpreting their data, in an imprecise manner e.g. *Sex-limited expression allows selection to modify the male size–trait reaction norm without constraining female growth pathways, whereas morph differentiation primarily alters allocation levels while maintaining a shared scaling framework* [L365]. What does 'allocation' mean developmentally, what is a 'scaling framework'? Similarly, "*differences in scaling architecture can reveal how developmental pathways partition exaggeration within a single species"* [L43]. What does partition mean? What is scaling architecture? There is an increasingly good understanding of the developmental mechanisms that regulate scaling, both in model organisms such as *D. melanogaster*, and non-model horned beetles. I encourage the authors to either limit their interpretation, or embed it in a more detailed review of the developmental literature.

5) It is not necessary to have two analyses. A single analysis of females, brachylabic and macropabic males is sufficient. Indeed, pooling all males together in a

comparison with females, when you know that the scaling relationship between forcep and body size is different between the male morphs, renders any inferences using the pooled data unreliable.

Minor points

6) The authors should present the formulae of the statistical models that they test in the MS. The results are challenging to interpret without them. The data are online, but currently hidden. I do not know whether the analysis scripts are similarly online but hidden, but should be made available.
7) [L317].“Although most empirical work has focused on mean trait expression, theory does not restrict condition dependence to shifts in intercepts”. I think a general interpretation of the condition dependence theory is that it pertains to the slope of scaling relationships, not just the intercepts (e.g. Bonduriansky, Am Nat, 2007, Vea et al, Am Nat, 2023).